

Nifty 100 Analytics

Analyst Guide

Covers the Streamlit dashboard, PDF tearsheets, and the REST API.

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1. Getting started

Set up the project once:

```
python -m venv .venv
.venv\Scripts\activate (Mac/Linux: source .venv/bin/activate)
pip install -r requirements.txt
```

Rebuild the database and every downstream file, in order:

```
python src/etl/loader.py
python -m src.analytics.run_ratio_engine
python -m src.screener.run_screener
python -m src.analytics.peer
python -m src.analytics.peer_report
python -m src.analytics.valuation
python -m src.analytics.clustering
```

2. The Streamlit dashboard - screen by screen

Start the dashboard with:

```
streamlit run src/dashboard/app.py
```

It opens at <http://localhost:8501>. The sidebar lists all 8 screens:

#	Screen	What it's for
1	Home	6 KPI tiles, sector donut chart, top 5 companies by score.
2	Company Profile	Search a ticker; see its KPIs, 10-year charts, pros and cons.
3	Screener	Filter the universe with sliders or the 6 presets; export to CSV.
4	Peer Comparison	Radar chart and side-by-side table for a peer group.
5	Trend Analysis	Overlay up to 3 metrics over 10 years, with YoY change.
6	Sector Analysis	Bubble chart (revenue vs ROE) and sector median KPIs.
7	Capital Allocation	Treemap of the 8 capital allocation patterns.
8	Annual Reports	Links to each company's annual reports by year.

3. Using the screener

The Screener screen (page 3) has two ways to filter companies:

- **Presets** - click one of the 6 buttons (Quality, Value, Growth, Dividend, Debt-Free, Turnaround) to auto-fill the sliders with that preset's thresholds.
- **Manual sliders** - 10 sliders in the sidebar for ROE, D/E, FCF, Revenue CAGR, PAT CAGR, OPM, P/E, P/B, Dividend Yield and Interest Coverage. Move any slider to change the result table live.

The result count is shown above the table (for example, "23 companies match your filters"). Click **Download CSV** to export the visible columns.

Banks and other Financials-sector companies are excluded from the D/E filter, because high leverage is normal for them. A "Debt Free" company always passes any minimum interest coverage filter.

4. Generating PDF tearsheets

Every company has a 2-page tearsheet PDF in reports/tearsheets/. To regenerate all of them (for example after a fresh data load):

```
python -m src.reports.tearsheet
```

Companies with no profit and loss data are skipped, and logged to output/skipped_tearsheets.csv. Companies with fewer than 3 years of data (for example JIOFIN) still get a tearsheet that carries a limited-history note. To regenerate the 11 sector PDFs and the portfolio summary PDF:

```
python -m src.reports.sector_report  
python -m src.reports.portfolio
```

You can also download a single tearsheet from the API - see section 5.

5. Calling the API

Start the API server:

```
uvicorn src.api.main:app --port 8000
```

Interactive documentation (try every endpoint from the browser) is at <http://localhost:8000/docs>. All endpoints are under `/api/v1`. Some examples:

```
curl http://localhost:8000/api/v1/health
curl http://localhost:8000/api/v1/companies/TCS
curl "http://localhost:8000/api/v1/screener?min_roe=15&max_de=1"
curl http://localhost:8000/api/v1/sectors
curl "http://localhost:8000/api/v1/peers/IT%20Services"
curl -o tcs_tearsheet.pdf http://localhost:8000/api/v1/companies/TCS/tearsheet
```

The full list of endpoints and their parameters is in `docs/openapi.json`, and can also be imported into Postman from `docs/postman_collection.json`.

6. Troubleshooting common issues

Problem	Likely cause / fix
"ModuleNotFoundError: No module named 'src'"	You are not in the project root. cd into the folder that contains src/, data/ and tests/
"ModuleNotFoundError: No module named 'pandas'" (or a similar)	Virtual environment is not active, or the libraries were never installed. Run pip install pandas
Streamlit says "no such table"	The database is empty or out of date. Run python src/etl/loader.py, then python src/etl/runner.py
A ticker shows "Ticker not found" on the Profile screen	Check the spelling against the Home screen's company list - tickers use the NSE ticker
The API returns HTTP 404 for a company	The ticker must be uppercase and match the id column in the companies table exactly
The API returns HTTP 400 on the screener	One of the numeric query parameters (min_roe, max_de, ...) is not a valid number
A tearsheet download 404s	That company has no profit and loss data and was skipped - check output/skipped_tickers
Streamlit and the API both need to run at the same time	Use two terminals: one running streamlit run src/dashboard/app.py, the other uvicorn src/api/main:app

7. Data notes and known limitations

A few things worth knowing before presenting numbers from this project:

- The database holds 10 broad sectors. The 11th sector report, Conglomerates / Other, is a cross-cutting group of conglomerate, holding-company and diversified businesses, whose companies also appear in their own broad-sector reports.
- Market data (stock prices, market cap, P/E, P/B, dividend yield) is SIMULATED for this project and should not be used for real trading or investment decisions.
- A handful of companies (for example BEL, HAL, INDIGO) show return-on-equity figures above 100% because of small or negative equity bases in the source data. These are flagged in output/outlier_report.csv and are capped (winsorised) in the composite quality score so they cannot dominate the screener rankings.
- 'Debt Free' companies (no interest expense) are treated as having infinite interest coverage everywhere in the project - the screener, the API, and the pros/cons generator.

8. Where each report file comes from

File	Generated by
output/screener_output.xlsx	python -m src.screener.run_screener
output/peer_comparison.xlsx	python -m src.analytics.peer_report
output/valuation_summary.xlsx	python -m src.analytics.valuation
output/cluster_labels.csv	python -m src.analytics.clustering
output/pros_cons_generated.csv	python -m src.nlp.pros_cons_generator
reports/tearsheets/*.pdf	python -m src.reports.tearsheet
docs/openapi.json	generated from the running FastAPI app (src/api/main.py)

9. Glossary

Term	Meaning
ROE	Return on Equity - net profit as a percentage of shareholder equity plus reserves.
ROCE	Return on Capital Employed - EBIT as a percentage of equity, reserves and borrowings.
D/E	Debt-to-Equity - total borrowings divided by equity plus reserves. 0 means no debt.
CAGR	Compound Annual Growth Rate - the smoothed yearly growth rate over a period.
FCF	Free Cash Flow - cash from operations plus cash used in investing.
ICR	Interest Coverage Ratio - operating profit and other income divided by interest cost.
OPM	Operating Profit Margin - operating profit as a percentage of sales.
P/E, P/B	Price-to-Earnings and Price-to-Book - valuation multiples from market price.
Composite score	A 0-100 score combining profitability, cash quality, growth and leverage.